YUANJUN (JOHNNY) DAI

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SUMMARY

AI Infrastructure and Applied AI Engineer, **PhD candidate** at CWRU. Specializes in the security, optimization, and observability of ML/LLM training and inference infrastructure. 8 research papers (7 first-author) at IEEE, ACM, and Springer; **2+ years of systems software engineering** at Cisco and Broadcom; 2+ years of **cross-functional** product ownership at Midea and Xiaomi.

EDUCATION

Ph.D. Candidate in Computer Science (AI Infrastructure) — 08/2019 – Sep. 2026 (exp.)
Case Western Reserve University

M.S. in Electrical and Computer Engineering (Computer Architecture)
University of Pittsburgh

B.S. in Electrical Engineering
Northeastern University

SKILLS

Languages	Python, C/C++
ML Frameworks	PyTorch (DDP/FSDP), vLLM, Megatron-LM, NCCL, BytePS, scikit-learn
Parallelism Paradigms	Data Parallel (DDP/FSDP), Model Parallel (TP/PP), Pipeline Parallel, Hybrid Parallel
Infrastructure	Kubernetes, Docker, CI/CD Pipelines, ArgoCD, Slurm, HPC
Systems & APIs	FastAPI, REST API, gRPC
Observability	eBPF (Kernel Tracing), Prometheus, Grafana, DCGM Exporter, Intel RAPL
Cloud & HPC	Lambda Labs HPC, Ohio Supercomputer Center, NSF FABRIC, CloudLab
Networking	SDN (Software-Defined Networking), P4, NPU/ASIC (Broadcom BCM SDK)

RESEARCH & SYSTEMS ENGINEERING EXPERIENCE

Research Assistant, AI Infrastructure & ML Systems — 08/2019 – Sep. 2026 (exp.)

Case Western Reserve University

Research focus: **security, optimization, and observability** of **ML/LLM** training and inference infrastructure.

DYNAMIX – Reinforcement Learning-Based Adaptive Batch Scheduling for Distributed ML | Paper / Code

- Designed a **PPO**-based **Reinforcement Learning (RL)** scheduler monitoring hardware signals (CPU/memory, bandwidth, retransmission rate via custom **eBPF** probes) and statistical signals (loss convergence, generalization) to dynamically adapt batch size
- Co-scales learning rate via **Linear Scaling Rule**; evaluated on **Ring AllReduce (DDP)** and **Parameter Server (BytePS)**
- Achieves **46% end-to-end training time reduction** across **8–64 A100 GPUs** with improved final model accuracy over fixed batch size baseline

PRISM – Priority-Based Selective Reliability for Distributed ML Training | Paper (Under Review)

- Designed a transport-layer library (**NCCL/Gloo → UDP**) hooking into **PyTorch DDP**'s AllReduce at bucket-ready events; classifies gradients by magnitude into high-priority (full **SACK**-based retransmission) and low-priority (local compensation via cached values)
- Supports kernel-bypass (**LibVMA**) and kernel-based (**GSO/GRO**) variants; achieves **8–15% throughput gain for CNNs and 15–21% for LLMs** over TCP, matching NDP latency at 144-worker scale

Scalable LLM Serving Infrastructure on Kubernetes | Code

- Independently designed and built end-to-end a **K8s** microservices platform (Gateway / Router / Worker) with **vLLM** inference
- **ArgoCD** GitOps, **Prometheus/Grafana** observability, and **CI/CD** pipelines across **50+ deployments**

Domain-Adapted LLMs for Clinical Risk Adjustment Coding | Paper

- Built a 115-label HCC multi-label classifier on highly imbalanced EHR data (MIMIC-V), improving long-tail performance via **inverse-frequency weighted binary cross-entropy loss** and **per-label threshold calibration**, achieving **0.775 macro-F1 / 0.742 micro-F1 / 0.9362 macro-AUC**.
- Trained **PubMedBERT** using **PyTorch DDP** on 2×NVIDIA RTX A6000 GPUs for efficient scaling, while fine-tuning **Qwen-7B** using distribution-aware **few-shot prompting** (5-shot with negative, frequent, and rare-label demonstrations) to enforce structured HCC outputs and improve sensitivity to sparse labels; scaled full-parameter training via **NeMo + Megatron-LM** (TP=1, PP=1) with **PyTorch FSDP (ZeRO-3)**, **FlashAttention**, and **activation checkpointing**, enabling training under GPU memory constraints (avoiding OOM).

Priority-Based eBPF Network Flow Telemetry for AI / DML Infrastructure | Paper / Code

- Designed PSketch, the first in-kernel priority-aware sketch using **eBPF**; high-priority flows tracked via in-kernel hashmap (O(1)), low-priority via **3-layer Count-Min Sketch**; enables **O(n) Top-K detection** with sub-linear memory, voting-based collision resolution and atomic operations for thread-safe updates
- Achieves **96% Top-K accuracy, 96.4% retransmission recall**, and <1.1% throughput overhead at 10 Gbps on FABRIC testbed

High-Resolution eBPF System Resource Profiling for DML Workloads | Paper / Code

- Designed eHashPipe, a dual-pipeline **eBPF** framework: (1) HashPipe-based memory profiler with cascading slot eviction and ptr2size deallocation tracking, (2) per-thread CPU tracker via sched_switch tracepoint
- Delivers **10–14× finer temporal resolution** than top/htop, ~11ms latency, and >90% Top-K fidelity on commodity Linux servers

Training-Time Side-Channel Attacks on Distributed ML Systems | Paper / Code

- Exposed MLaaS training-phase side-channel vulnerability via deep analysis of DML computation/communication behavior in Parameter Server architecture
- Applied **K-means clustering** on push/pull traffic (silent period detection), **Fourier transform** on **Intel RAPL** for noise reduction, **Decision Tree** for layer-type classification, and **polynomial regression** for **activation function profiling**
- Achieves **>99% layer-type accuracy and <1.5% tensor-size error** across ZFNet, AlexNet, and VGG

Scotch – Elastic SDN Overlay for Control-Plane Scalability under DDoS | Paper / Code

- Designed an **OvS**-based elastic overlay: when physical switch OFA is saturated, tunnels new flows to a **vSwitch** pool via **event-driven** Packet-In Edge messages
- Separates small DDoS flows (terminated at vSwitches) from large flows (migrated to physical paths); achieves **>70% controller congestion reduction** on 34-node GENI Clos topology

INDUSTRY EXPERIENCE

Software Engineer II
06/2015 – 08/2016

Cisco Systems, Inc.

- Developed NPU/ASIC features for a 100 Gbps core router (Azure, AWS, Alibaba Cloud) using Broadcom BCM SDK.
- Contributed to QoS and ECMP modules, reducing forwarding latency under heavy load.

Software Engineer Intern
10/2014 – 05/2015

Broadcom (Emulex)

- Replaced polling-based data acquisition with a DMA + circular-buffer architecture in an RTOS pipeline, **reducing CPU load by 30%** and improving sampling rate and control-loop responsiveness.
- Ported a Linux-only management CLI for BladeEngine ASIC-based CNA hardware to Windows Server, adapting system calls, threading models, and error handling.

Product Manager
09/2016 – 06/2018

Midea Group

- Served as technical PM for **Fun Oven II**, a multi-disciplinary smart appliance spanning data collection, ML model training & deployment, mobile app integration, cloud backend, and culinary algorithm design.
- Participated in resolving technical challenges across all dimensions — from training AI image recognition models and deploying Docker services on Alibaba Cloud, to defining cooking curves and app UX.
- Product debuted at **2018 AWE Show** and won the **Red Dot Design Award** · Red Dot ↗ · PR Newswire ↗

Technical Product Manager
07/2018 – 01/2019

Xiaomi

- Market-facing PM role: conducted competitive analysis and user research to define product requirements for a smart IoT appliance.
- Assessed technical feasibility and complexity of proposed features; delivered initial technical proposals covering architecture trade-offs and implementation risks to align engineering and business stakeholders.

PUBLICATIONS

8 papers total, 7 first-author, 5 published at IEEE CCNC, IEEE ICNC, ACM TOIT, and Springer SecureComm; 3 under review (incl. ACM TOPS). Full list: johnny-dai-git.github.io/portfolio